

NOVARA // AI Polar Energy Management System (PolarOPS)

Target Agency: Ministry of Earth Sciences (MoES) & National Centre for Polar and Ocean Research (NCPOR)
Problem Statement ID: 26061 · Indian Antarctic Research Bases: Bharati & Maitri
Prepared Under: Ministry of Education's Innovation Cell (MIC) & AICTE Post-SIH Guidelines

1. Executive Summary & Commissioning Scope

In accordance with **Guideline #1** of the Ministry of Education's Innovation Cell (MIC), hackathon projects are operational prototypes that require systematic engineering, field telemetry integration, and industrial hardening before live station commissioning. This document constitutes the formal **6-Month Detailed Project Plan** for transitioning the **NOVARA (PolarOPS)** operational prototype into a field-deployed, cyber-secured Energy Management System for **Bharati Station** (Larsemann Hills) and **Maitri Station** (Schirmacher Oasis).

Core Operational Deliverables & ROI Targets:

- **Direct Fuel Abatement:** Achieve audited annual savings of **42,600 Liters** of polar aviation turbine fuel (ATF-50), yielding **₹83.07 Lakhs** in direct fuel purchase savings.
- **Airlift Logistics Economy:** Eliminate **14 Kamov Ka-32 / Bell 412 heavy helicopter sling sorties**, securing **₹63.0 Lakhs** in avoided charter flight costs (Total Expedition Savings: **₹1.46 Crores / year**).
- **Zero Blackout Guarantee:** Real-time sub-second Battery Energy Storage (BESS) synthetic inertia ($\leq 10\text{ ms}$ transient response) with Automatic Transfer Switch (ATS) genset failover.
- **Antarctic Treaty Madrid Protocol:** Abatement of **114.2 Tonnes of CO₂** and elimination of unburnt hydrocarbon soot near Adélie penguin and snow petrel sanctuaries.

2. 6-Month Phased Implementation Roadmap

The project is structured into four sequential phases over a 26-week timeline, adhering to MIC Guidelines #2, #4, and #8 with bi-weekly milestone reviews with NCPOR technical mentors.

Phase & Timeline	Work Package & Core Activities	Key Deliverables	Verification Gate
Phase 1 Weeks 1–4 (Month 1)	Telemetry Schema & NCPOR Baseline Calibration • Visit NCPOR HQ (Goa) for baseline data ingestion. • Audit electrical single-line diagrams of Bharati & Maitri. • Calibrate Volvo Penta 120kW & Deutz genset consumption curves.	• Site Telemetry Interface Spec (TIS). • Calibrated Station Hardware Profiles. • Initial Data Security Plan.	NCPOR Mentor Sign-off on Baseline Telemetry Registry

Phase 2 Weeks 5–12 (Months 2–3)	Hardware-in-the-Loop (HIL) & SCADA Gateway • Deploy ruggedized fanless Industrial Edge PC. • Interface Moxa IEC-61850 / Modbus-TCP gateway. • Connect real-time test bench for 48 field registers (voltage, freq, temp, oil pressure).	• Operating HIL Substation Simulator. • HiGHS MILP 24-hr Rolling Solver. • Sub-40ms unit commitment loop.	Substation Fieldbus Latency Benchmark (<120ms)
Phase 3 Weeks 13–20 (Months 4–5)	Thermal Co-Gen Coupling & Cybersecurity • Integrate Combined Heat & Power (CHP) heat-loop sensors ($\dot{Q} = \dot{m} C_p \Delta T$). • Enforce deterministic safety guardrails. • Complete CERT-In compliance & vulnerability assessment. • Implement Commander PIN / RBAC authentication.	• CHP Thermal Dispatch Module. • CERT-In Security Clearance Audit. • Offline Satellite VSAT Sync Mode.	Third-Party Cybersecurity Audit Certification
Phase 4 Weeks 21–26 (Month 6)	Final Acceptance Testing & Commissioning • Pre-deployment staging on charter vessel <i>MV Vasily Golovnin</i>. • Factory Acceptance Testing (FAT) with NCPOR engineering cell. • Station Commander operational manual & training.	• Packaged Edge Runtime Artifacts. • SOP Operational Logbook. • Final Project Completion Report.	Formal Handover to MoES / NCPOR Directorate

3. Hardware & Commercial Software Procurement Plan

Per **MIC Guideline #5** (*Ministry should procure commercial software/hardware critical for development and implementation*), the following equipment is required for the Hardware-in-the-Loop test bench and field deployment:

Item & Specification	Qty	Operational Purpose	Est. Cost (₹)
Advantech UNO-2271G Industrial Fanless PC Intel Core i5, -40°C to +70°C wide temp, DIN-rail, 24V DC	2 Units	Polar station on-premise edge computing host for HiGHS MILP optimizer and local SQLite database.	■1,85,000
Moxa NPort IA5450A / MGate 5105 Gateway IEC-61850 MMS to Modbus-TCP/RTU, 4-port serial isolation	2 Units	Fieldbus gateway interfacing genset ECUs and BESS inverters with 120ms polling heartbeat.	■1,30,000
Schneider Electric PM8000 Class 0.2S Power Meters True RMS, 63rd harmonic capture, Modbus-TCP	4 Units	Substation bus power quality monitoring, transient frequency tracking, and reverse power protection.	■1,60,000
Industrial PT100 RTD Temperature Transmitters 4-20mA loop powered, -60°C to +150°C cryogenic rated	8 Units	Monitoring battery cell core temperatures and engine jacket coolant heat-recovery glycol loops.	■48,000
Open-Source Software & Tooling HiGHS MILP, FastAPI, PostgreSQL/TimescaleDB, Docker	Enterprise	Core algorithmic dispatch engine, digital twin simulator, and deterministic guardrail suite.	■0 (Open Source)

Total Hardware & Software Outlay	-	Procured directly by Ministry / NCPOR under SIH Rule #5	■5,23,000
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4. Expert Mentorship & Institutional Framework

In accordance with **Guidelines #6, #7, #9, #12, and #13**, project development will be supervised by a multi-disciplinary technical advisory board:

- **NCPOR Technical Mentor (Guideline #7):** A designated senior polar electrical/mechanical engineer from NCPOR to guide station single-line topology, diesel governor response, and fast-ice logistics.
- **Cybersecurity Auditor (Guideline #12):** Engagement of a CERT-In certified cybersecurity specialist to perform penetration testing, verify air-gapped substation isolation, and secure WebSocket channels.
- **Industrial Product Design Mentor (Guideline #13):** Mechanical advisor for DIN-rail enclosure thermal dissipation and vibration dampening during Antarctic transit.
- **Institutional Faculty Mentor (Guideline #9 & #10):** College faculty supervisor coordinating remote laboratory facilities, student academic leave, and college support without any financial burden on the educational institution.

5. Team Stipend & Travel Budgetary Outlay

Structured in strict compliance with **MIC Guideline #14** (Monthly student stipend of ■10,000–■15,000 for up to 6 members) and **Guideline #11** (Travel, Stay & Field Survey Allowances):

Expenditure Head	Calculation Basis & Entitlement Norms	Total Amount (■)
Student R&D Stipend (6 Months) (SIH Guideline #14)	6 student team members × ■15,000 / month × 6 months (Direct disbursement to students by Ministry)	■5,40,000
Long-Distance Travel Allowance (SIH Guideline #11)	AC-III Tier train fare for team visits to NCPOR HQ (Goa) · 2 site visits × 6 members (Reimbursed on actual bills)	■72,000
Daily Stay & Lodging Allowance (SIH Guideline #11)	■1,500 / day × 6 members × 8 travel days total across integration workshops	■72,000
Local Commute & Field Survey (SIH Guideline #11)	■1,000 local travel / day + ■500 field survey allowance / day × 6 members × 8 days	■72,000
Consolidated Human Capital & Travel Grant	Allocated per Ministry of Education Innovation Cell Guidelines	■7,56,000

6. Intellectual Property (IP) & Commercialization Terms

As formalized in the **SIH Intellectual Property (IP) Charter**:

1. **Student Ownership:** The Intellectual Property rights of the NOVARA algorithmic microgrid optimizer, digital twin mathematical formulations, and software architectures reside with the student development team.
2. **Government Royalty-Free Access:** The Ministry of Earth Sciences (MoES) and NCPOR are granted a perpetual, irrevocable, royalty-free license to deploy, operate, and maintain the solution at all Indian Antarctic (Maitri, Bharati) and Arctic (Himadri) scientific stations.
3. **Startup Incubation:** The student team is committed to incubating a registered Clean-Tech startup post-deployment, enabling scaling of this polar-grade microgrid resilience technology to Indian defense microgrids (Siachen, Ladakh border outposts) and

remote Himalayan civilian settlements.

Submitted By:

Lead Developer & Student Representative
Team NOVARA (PolarOPS)
Smart India Hackathon 2025

Acknowledged By:

Institutional Faculty Mentor
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Engineering
(Educational Institution Endorsement)

Approved For Initiation:

Directorate of Polar Science / IT Cell
Ministry of Earth Sciences (MoES) /
NCPOR
Government of India